ZAIO

Full-Stack AI Engineer Bootcamp - The Handbook

2025

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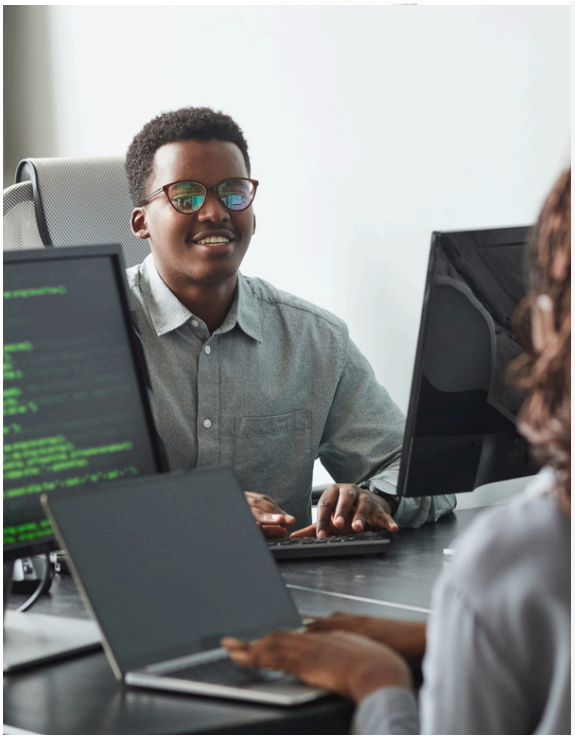
WELCOME

Over 200,000 technical roles are left unfilled every year in South Africa.

The opportunity is extremely massive!

We are thrilled to have you join us on this transformative journey into the world of coding. This handbook is your compass, providing essential information, guidelines, and insights to ensure your experience with us is enriching and successful.

Throughout the pages ahead, you will find details about the curriculum, key dates, support systems, and the exciting opportunities that await you. Our commitment is not just to teach you how to code but to empower you with the skills and knowledge needed to thrive in the dynamic tech industry.



> Entry-level positions start at R 390 000 per year, while most experienced workers make up to R 7 200 000 per year.
>
> -
>
> Talent.com

OUT COMES

At Zaio, our primary goal is to equip you with the skills and knowledge you need to excel in the field of programming. By the end of this bootcamp, you can expect to:

Master Programming Fundamentals: Develop a solid foundation in programming languages, algorithms, data structures, and problem-solving techniques.

Build Real-World Agentic Projects: Gain hands-on experience by working on projects that simulate real industry scenarios, allowing you to showcase your abilities to potential employers.

Collaborate Effectively: Learn how to work efficiently in a team, participate in collaborative coding environments, and communicate your ideas effectively.

Receive Career Support: Benefit from resume building, interview preparation, and job placement assistance to help you launch your career in tech.

We are committed to your success and will do everything we can to support you throughout your journey with us.

HARDWARE REQUIREMENTS

To effectively participate in a full-stack web development course, it's essential to have a laptop that meets certain minimum hardware requirements to ensure smooth operation of development tools and environments. Here are the recommended specifications:

Operating System:

- A current Windows-based laptop with PowerShell or a Mac with Terminal.

Processor:

- A dual-core Intel i5, AMD 3000+ (CPU), or Apple M1 processor, or greater in power.

Memory (RAM):

- At least 4 or 8 GB of RAM is required for efficient multitasking and handling of development environments.

Storage:

- A solid-state drive (SSD) with at least 256 GB of storage is recommended for faster data access and improved performance.

Internet Connection:

- For online classes, a high-speed internet connection is essential. Recommended minimum speeds are 10 Mbps for download and 3 Mbps for upload.

If you are looking to buy a laptop and are confused - please email asif@zaio.io for guidance.

THE SPRINTS

In this transformative coding journey, we'll be embarking on a series of 30-week sprints. Each sprint is a dedicated step forward in your learning and growth. Get ready to dive into the world of coding, challenge yourself, and achieve significant milestones

Weeks	Topics covered	Projects	Outcomes
01 - 10	• Orientation • HTML • CSS • Git • JavaScript	• Tesla landing page • Netflix landing page • Youtube home page • Twitter clone • Google keep app	• Build your own HTML and CSS pages or websites • Problem solve using programming • Make your webpages interactive
11 - 15	• ReactJS • Redux • Routing • API calls • Advanced React • Context • Hooks	• Google keep app • Airbnb clone • Amazon clone • TicTacToe game	• Build dynamic web applications using ReactJS • Connect to third party APIs
16- 20	• NodeJS • NPM • Express • RESTAPIs • MongoDB • Postman • Async & Await	• E-commerce app • Food Delivery app	• Connect your app to a database • Query databases • Connect your backend to third party APIs • Build your own REST APIs
21 - 24	• Cursor AI for vibe coding and building apps faster • Python and OOP • Cap Stone project	**Project will be revealed on the 20th week**	• Problem solve using Python • Build a fully working ReactJS + NodeJS (MERN Stack) app
25 - 30	• Building AI agents with N8n • Building AI agents with Langchain • Full Stack Agentic applications	**Make your CapStone Project an Agentic Fullstack App**	• Build AI agent prototypes using no-code • Build complex agents using langchain • Add these agents to your fullstack apps

KEY DATES

The bootcamp is scheduled to run from the <DATE> for 7 months.

Please take note of the following important dates:

Orientation Day: 27 May 2025 , 10 am

Topic: May Bootcamp Orientation

Orientation Day Link: <TO BE ADDED>

Meeting ID: 867 2332 9016
Passcode: 696630

Virtual Graduation Day: (Time and Date to be confirmed)

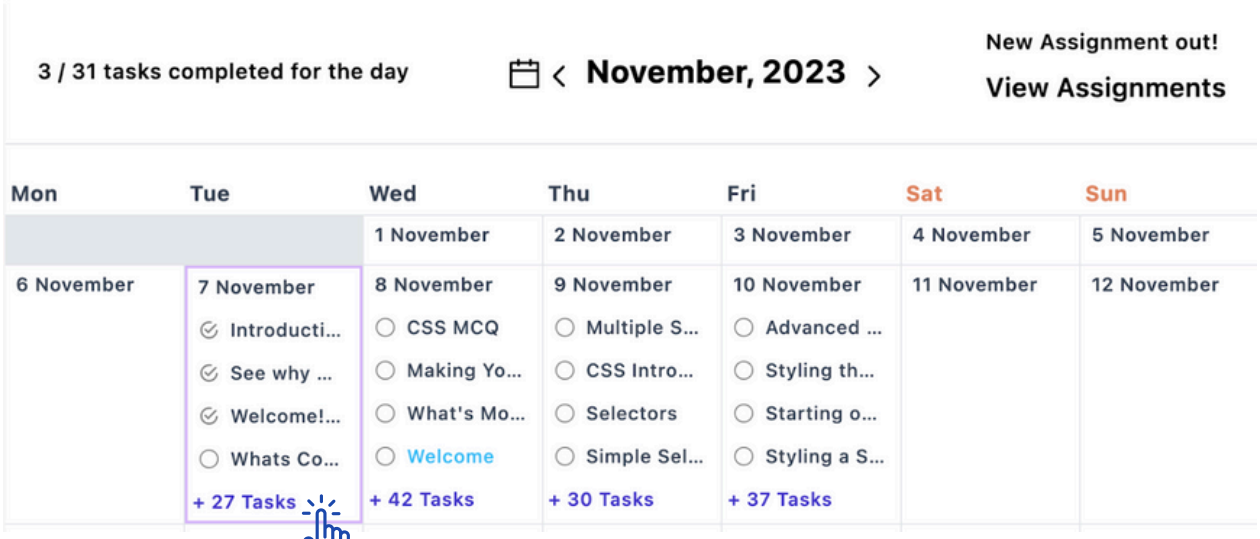
HOW THIS BOOTCAMP WILL WORK

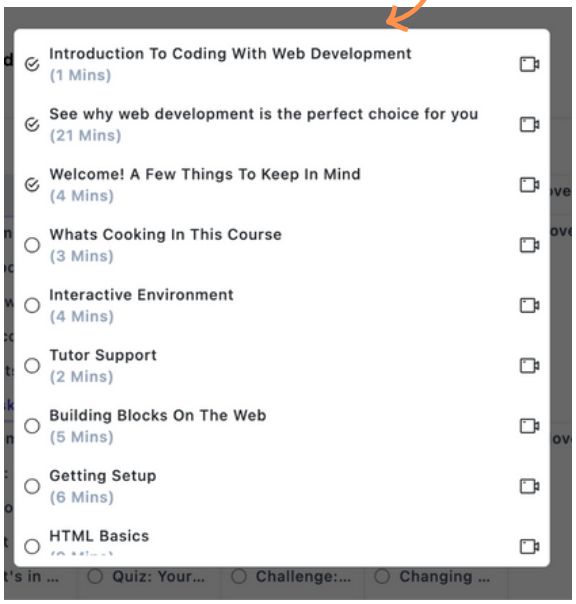
As a distinguished participant in this program, your enrolment grants you access to our advanced Learning Management System.

This sophisticated platform has been curated to offer a comprehensive suite of resources and interactive modules, strategically designed to elevate your proficiency in full-stack AI engineering.

Each student will have access to a calendar that has daily tasks setup for them. You will need to complete all the tasks on time to make sure you complete the bootcamp in 7.5 months.

Below is an example what your calendar will look like on your dashboard.





Daily tasks view when you zoom into each calendar tile

In our commitment to fostering a disciplined and results-driven learning environment, your calendar serves as a visual representation of your progress. Here's a breakdown of the color-coded system:

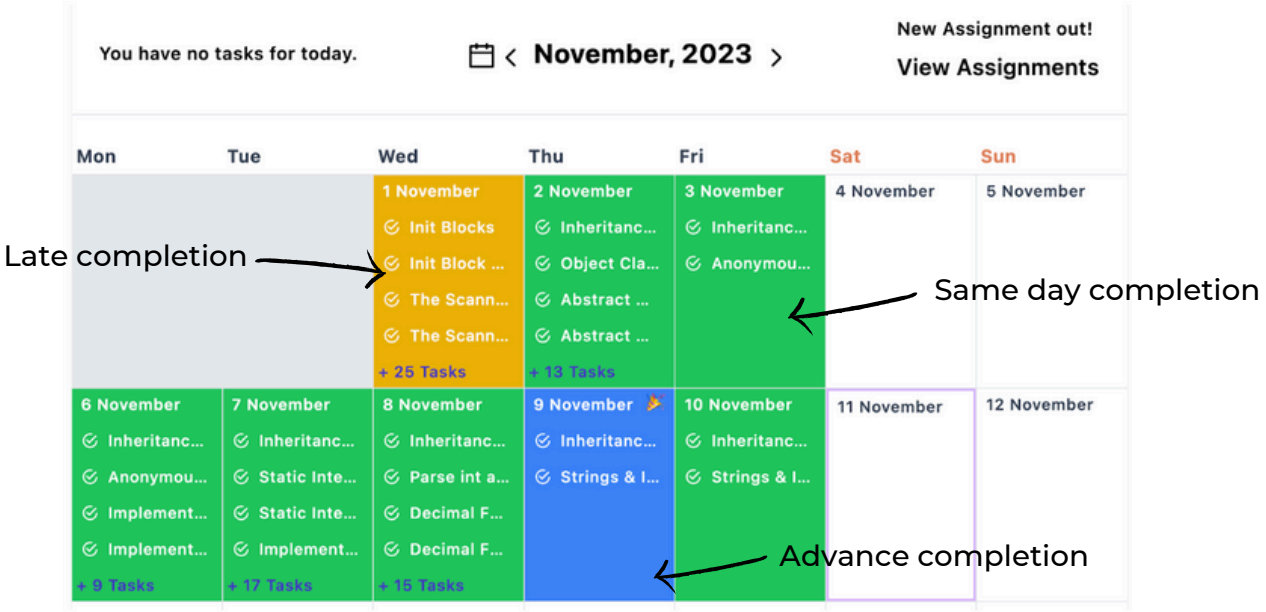
- **Green Tiles (Same Day Completion):** Demonstrates your ability to meet deadlines promptly, showcasing a strong work ethic and commitment to timely task completion.

- **Yellow Tiles (Late Completion):** Indicates completion after the designated day or more. While not ideal, it serves as a valuable learning opportunity to enhance time management skills.

- **Blue Tiles (Advance Completion - One Day or More):** Reflects proactive planning and efficiency, showcasing your ability to work ahead of schedule.

- **Red Tiles (Incomplete):** Highlights tasks that remain unfinished for the day, signaling a need for attention and completion. Something you need to avoid.

The cumulative effect of green and blue tiles not only signifies your dedication but also instills confidence in our team regarding your capability to meet deadlines consistently. This metric plays a pivotal role in our assessment and placement processes.

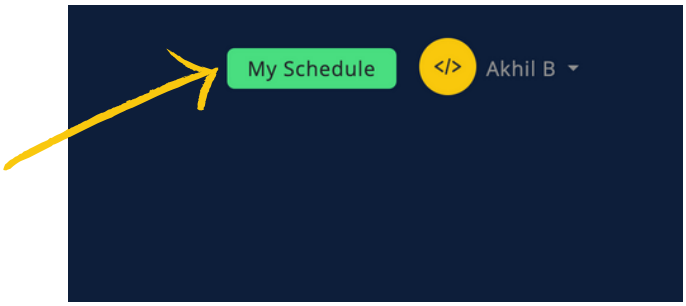


HOW TO GET ACCESS TO YOUR CALENDER

Make an account on www.zaio.io and add the email address you used to create the account on this google sheet, in the May-25 BC sheet. (https://docs.google.com/spreadsheets/d/1SUZWT-b18BioOSX5s0sZHtRFws31DHXmQfNmrtw8NaQ/edit?usp=sharing).

We will enrol into the course and create the calendar for you.

When your calendar is ready you will see a green ‘My schedule’ button on the top right of your dashboard.



Make sure your email address is on the sheet by Tuesday, 27 May 2025, 1 pm.

LIVE CLASSES

We will have live classes every **9 am - 11 am** on **Tuesdays** and **6 pm to 8 pm** on **Thursdays** for the first 12 weeks of the bootcamp which will then reduce to 1 live class every Monday at **9 am - 11 am.**

All live classes will be recorded. For the students who are not able to attend a specific live classes and studying at another time, we will have tutors available on standby to assist when you have any doubts.

These live classes will be used to recap what was covered during the week on the LMS, make sure you complete your work before coming to class.

HOW TO JOIN CLASSES

Tuesday Morning Sessions (9 am - 11 am):

Topic: Tuesday Live Session

Time: Jun 3, 2025 09:00 AM
Every week on Tuesday

Join Zoom Meeting

Meeting ID: 859 2569 3051
Passcode: 936862

Thursday evening sessions (6 pm - 8 pm)

Topic: Thursday Live Session
Time: Jun 5, 2025 06:00 PM
Every week on Thursday

Join Zoom Meeting

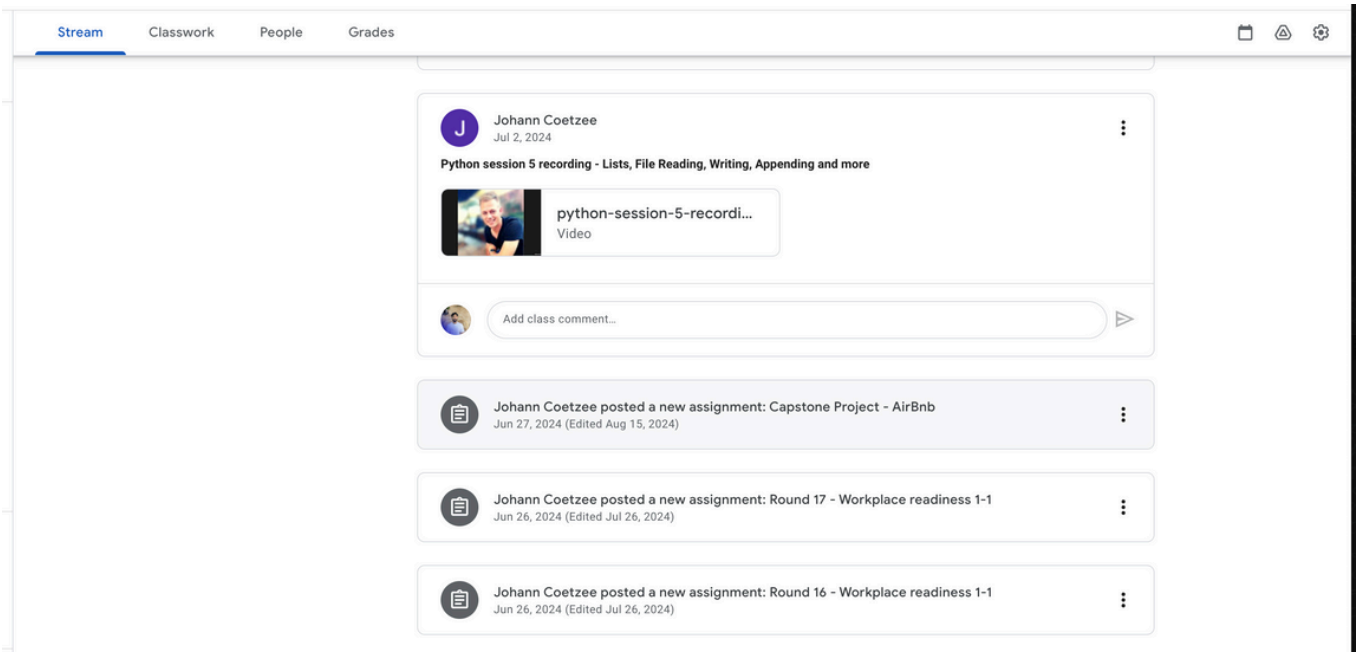
Meeting ID: 873 8947 0102
Passcode: 463393

GOOGLE CLASSROOM

We will be using Google Classroom as the main LMS to post all recorded lectures.

Assignments will also be shared on Google Classroom and they will have to be submitted on Google classroom and they will be marked by your allocated Tutor.

You will be added to Google classroom after the orientation meeting - an invitation link will be send to your email address and you will have to join that class room using the link.



GRADE BREAKDOWN

This is how we will determine the final grade, based on the following areas.

- Final Project (React + Node) (35%):
  - The final project serves as a comprehensive evaluation of the participants' ability to apply their skills to real-world scenarios. In this case, they will be working on a project using React and Node, demonstrating their proficiency in full-stack development.
- Assignments (25%):
  - Assignments gauge the participants' ability to independently apply the concepts learned throughout the BootCamp. These tasks may involve hands-on coding, problem-solving, and project-based assignments to reinforce their understanding of the course material.
- Coding Challenges (25%):
  - Coding challenges focus on assessing the participants' problem-solving skills and coding proficiency. These challenges may cover a range of topics, encouraging participants to think critically, write efficient code, and demonstrate their mastery of various programming concepts. Marked on Zaio platform.
- Multiple-Choice Questions (MCQs) (15%):
  - MCQs provide a structured approach to evaluating participants' theoretical knowledge and conceptual understanding of the course material. These questions may cover a broad spectrum of topics, ensuring a comprehensive assessment of their theoretical foundation. Marked on Zaio platform.

THE TEAM



Kago Kola
Lead Instructor



King Robert Kehnide
Teaching Assistant



Dave Fortune
Teaching Assistant



Abi Faith
Admissions
abi@zaio.io



Asif Hassam
Technical support



Suhana Patel
Student Success

FEES

We understand the importance of financial planning, and we aim to make the payment process as seamless as possible. Please take note of the following information regarding fees for the Bootcamp:

Total Fees: **R 38,950**

Payment Deadline: Fees must be paid in full before the orientation day, which is scheduled for 27 May 2025.

Payment Options:

Option 1: **Upfront Payment**

You can pay the full amount upfront using the Paystack link: https://paystack.com/pay/zaio-bootcamp. This secure online payment platform ensures a quick and straightforward transaction.

Option 2: **Financing with Capitec and Manati**

If you prefer a financing option, we have partnered with 2 partners (Capitec and Manati) to facilitate the process of securing a study loan. They assist numerous students in obtaining loans, making education more accessible.

If you choose this option, it is crucial to respond promptly to communications, keep a close eye on your emails, and cooperate with Capitec throughout the process.

Notification from Financing Partners

If you opt for Capitec education financing, you should expect to be contacted within 24 hours of receiving your acceptance letter. This communication will guide you through the steps of securing a study loan.If you have not been contacted by Capitec, please email: **abi@zaio.io,** Should your Capitec loan fail, we will then take your application to Manati who are more lenient when giving out study loans.

We understand that managing finances is a significant aspect of your education journey. If you encounter any challenges or have specific questions regarding fees and payment, please do not hesitate to reach out to **abi@zaio.io**. We are here to assist you in ensuring a smooth and successful payment process.

Remember, your commitment to payment by the specified deadline is crucial to securing your place in the bootcamp and ensuring a positive start to your learning experience.

COMMUNICATION CHANNELS

As we gear up for an exciting coding journey together, we want to keep you informed about our communication channels.

Before Orientation Day:
Until the official start of the boot camp, all communication will take place via email. Please ensure that you regularly check your email inbox for important updates, announcements, and any pre-boot camp information.

After Orientation Day:
Once the boot camp officially begins, all ongoing communication will transition to our Discord channel. Discord will be our central hub for discussions, announcements, and real-time interaction. Make sure to join our Discord server to stay connected with your fellow participants and our team.

We understand the importance of staying connected, so please keep an eye on your emails until the boot camp officially commences. After that, Discord will be the primary platform for all communication.

DISCORD CHANNEL

To ensure a smooth onboarding process and effective communication throughout the program, we've set up a dedicated Discord community. Please join our Discord server using the following invite link: https://discord.gg/hHzyG4FsU8

Once you're in, ensure your name shows as "**Name from May25 BC**", for example **"Mpho from May25 BC"**. This will help us know that you are from the May Bootcamp cohort at all times, and add a nice display picture too! Feel free to continue and introduce yourself in the "introduce-yourself" channel. The introduction is crucial because we will add you to the private bootcamp channel based on seeing you on discord.

TUTOR SUPPORT

Discord Tutoring:

Have a coding question? Simply post it in the designated Discord channel during our boot camp. A tutor will be promptly assigned to assist you. Look out for the Discord invite during the orientation session to get started.

Tutor allocation:

Following your orientation, we are pleased to inform you that a dedicated tutor will be assigned to support you throughout the bootcamp. Your tutor will be Robert or Dave, and they will remain your primary point of contact until the program concludes.

The role of your tutor is to motivate and assist you in completing your assignments and projects. They are committed to your success and may occasionally be assertive to ensure you stay on track with your tasks. Please understand that this approach is intended for your benefit. We kindly ask that you treat your tutor with respect, as they are dedicated to helping you achieve your goals.

Should you feel at any point of the bootcamp that you want to change your tutor - please let Suhana (Student success manager) know.

Support Hours:

Tutors will host support hours every

Tuesdays 2pm to 4 pm and **6 pm to 8 pm**
and
Thursdays at **10am to 12 pm** and **6 pm to 8 pm**

During support hours, you can book a video call with a tutor using the Calendly links that will be provided during orientation and on the discord channel. Alternatively, you can book a tutor directly from the ZAIO dashboard.

We're here to support your learning journey. Get ready for personalized assistance and collaborative coding sessions!

TEAM WORK CERTIFICATES

On discord we have a levelling system. This system allows us to gauge who are most engaging individual in the community and gives us a sense of your ability to help your peers and be a good team member. This plays a huge part of your soft skills and we are able to confident say 'Yes' to companies who are looking to hire from us when they ask us about your ability to work in a team.

Here are the various tags you can attain :

Lvl 1 Code Explorer – Just starting out and exploring the basics.
Lvl 2 Curious Coder – Showing interest and asking thoughtful questions.
Lvl 4 Helper – Actively helping others by answering questions.
Lvl 6 Team Player – Demonstrating good teamwork and collaboration.
Lvl 9 Mentor in Training – Starting to mentor and guide others in coding.
Lvl 12 Team Player Pro – Working well in teams and contributing to group
Lvl 15 Code Mentor – Actively mentoring others and sharing knowledge.
Lvl 18 Team Player Ultra Pro Max - Highest level of Team player category - a must hire.
Lvl 23 Code Mentor Pro - Great Coding mentor
Lvl 30 Code Mentor Pro Max - Ability to be a Senior Dev One day

You will be holding only one tag(role) at a time. e.g the moment you level up from **Code Explorer** to **Curious Coder** - your new tag(role) will be **Curious Coder** and the **Code Explorer** tag will fall away.

To know your roles in a huge server easily, you just have to search for a comment of your own (top left) and click on your profile picture. e.g



You will get XP points for engaging and answering questions/ helping students in the discord server. The XP points will be based on the effort of the response you are providing. You will also get XP points for asking questions. The more time you spend in framing your question clearly the more points you will get. Same for answers.

To know how many XP points your are from the next level type type **/rank** in the **#rank-commands** channel. This command will not work else where.

The **#who-leveled-up** channel will show you who levelled up and earned a role based on their contribution to this community.

Should you get to any of that tags listed below - you can DM me and I will provide you with a certificate of participation for each tag.

- Team Player Pro
- Team Player Ultra Pro Max
- Code Mentor Pro
- Code Mentor Pro Max

YOUR FINAL PORTFOLIO

Our primary objective is to empower you with the skills and experience to build an outstanding portfolio by the end of this boot camp. Throughout the program, you'll engage with multiple projects, spanning various tech stacks including HTML, CSS, JavaScript, ReactJS, and NodeJS.

To provide you with a glimpse of the impactful portfolios our previous students have crafted after completing our courses, here are some examples:

https://portfolio006.netlify.app
https://km-bootstrap-portfolio.netlify.app
https://nkanyiso-portfolio.netlify.app
https://lsm-bootstrap.netlify.app
https://ndumiso-portfolio.netlify.app

As you progress through this boot camp, remember to consistently update your portfolio after completing each project. Your portfolio is more than just a showcase; it's your strongest asset when venturing into the job market for the first time.

Make each project count, and let your evolving portfolio be a testament to your growth and expertise. We're here to support you in creating a portfolio that speaks volumes about your skills and potential.

PLACEMENT & CAREER

As you approach the end of our bootcamp handbook, we want to share the exciting prospects that await you.

Zaio is committed to supporting students in securing placements within our network of partner companies, offering opportunities for Junior Developer roles or paid internships. Achieving a final mark of 70% and above significantly enhances your chances of placement.

Paid internships go for 12 months, providing a monthly salary of **R4000**. Junior Developer full-time roles offer competitive salaries ranging from **R10,000 to R15,000** per month.

The focus in your initial year should not solely be on the salary; it's about establishing yourself in the tech industry. Exponential salary growth is anticipated from the second year onwards.

Your primary objective should be accumulating experience at tech companies, setting the stage for sustained career growth.

Refer to the graph below from Offerzen, a prominent tech recruitment company based in Cape Town, illustrating the correlation between salary and experience. It emphasizes the importance of gaining experience for long-term career success.



Your journey beyond the bootcamp is one of continual growth. Embrace the opportunities ahead and let your tech career flourish.

EXCITING TIMES AWAIT—LET'S EMBARK ON THIS JOURNEY TOGETHER!

CONTACT

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